

Adjoint Sensitivities and Gradient-Based Manning Roughness Calibration in RDycore

DRAFT for internal review

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Abstract

We present a discrete-adjoint capability for RDycore, the exascale river dynamical core of the Energy Exascale Earth System Model (E3SM), and use it for gradient-based calibration of spatially distributed Manning roughness. An exact assembled Jacobian of the finite-volume shallow-water right-hand side — closed-form Roe-flux blocks by hand-written forward-mode differentiation, with source-term and boundary-condition blocks — drives adjoint sensitivities through PETSc’s `TSAdjoint`, bound-constrained calibration with TAO, and fully implicit time stepping; one gradient costs about one extra forward solve, independent of the parameter count, and every derivative is validated against finite differences of the full nonlinear solve. Implicit time stepping, which the same Jacobian makes practical, also proves necessary: the differentiable form of Manning drag is friction-stiff, and explicit stepping is unstable at operational time steps. Twin calibrations — which use model output from a known synthetic roughness field as the observations, so that recovery can be scored against the truth — show that whether the field can be recovered at all depends on the observing geometry and on the observable, not on observation counts: surveyed high-water marks recover the land-cover roughness classes covering 83% of a 2.93M-cell Hurricane Harvey domain, where the real stream-gauge network recovers almost none. Against the real survey, the same evaluation isolates a downstream region the model cannot drain — a defect no roughness value can absorb, and a finding in its own right — and over the rest of the domain the model, run with textbook roughness values looked up from the land-cover table and no calibration at all, errs by 1.51 m in mean peak water-surface elevation. This is, to our knowledge, the first *transient* discrete adjoint in an E3SM component (the land-ice component inverts a *steady* solve), obtained by differentiating the production solver in place — its discretization, data structures, and GPU backends — rather than rewriting the model in a differentiable-programming framework.

1 Introduction

Manning roughness is the dominant tunable parameter of depth-averaged flood models, and its calibration is traditionally manual or ensemble based. Adjoint methods deliver the gradient of an observation misfit with respect to *every* cell’s roughness at the cost of about one extra forward solve, making distributed calibration tractable. The approach is well established in shallow-water modeling, from channel networks [1] and unstructured finite-volume twin experiments [2] to GPU-accelerated optimizers [3]. It is deployed in variational-assimilation platforms such as Dass-Flow [4, 5, 6] and the Thetis coastal model, which calibrated a spatially varying Manning field for the Baltic Sea with an automatically generated adjoint [7], following storm-surge sensitivity studies that addressed the overfitting inherent in distributed friction estimation [8]. The closest published work to ours is Pujol et al. [9], who infer distributed friction in a 2D urban flood model by variational assimilation of multi-source data *including high-water marks*, and use adjoint-derived global sensitivity measures to decide how to spatialize the friction field before assimilating — the same observable and the same design move that Sections 7.2 and 7.3 employ, at street scale and with a Tapenade-generated adjoint of a solver built for assimilation. What we add is neither the adjoint nor the observable but the setting and the question: a production Earth-system component differentiated in place, and a measurement of how much of the model’s error the parameter can explain at all.

A recent generation of *differentiable* flood and hydrology codes obtains gradients by rewriting simplified solvers inside machine-learning frameworks: Inunda (PyTorch, local-inertial, calibrated on a Hurricane Harvey hindcast) [10], Hydrograd (Julia) [11], AegirJAX [12], CaMa-Flood-GPU [13], and differentiable routing [14]. These demonstrate demand for exactly this capability, but none is a production Earth-system component: they trade the validated numerics, unstructured meshes, and coupling infrastructure of codes like RDycore for differentiability.

This paper takes the complementary route: make the production code differentiable. That route has a long precedent outside E3SM — most prominently MITgcm, whose exact adjoint is generated by automatic differentiation of the production ocean model [15] and underpins two decades of ECCO state and parameter estimation [16]. Ours differs in how the derivative is obtained rather than in ambition: the Jacobian is hand-differentiated and assembled, which makes it available to the implicit solver as well as to the adjoint, and Section 6 argues that for a differentiable friction term the implicit solve is not optional. Inside E3SM the precedent is narrower: the land-ice component MALI [17] routinely inverts for basal friction with an adjoint obtained by automatic differentiation of its Albany velocity solver [18] — but that is the adjoint of a *steady* nonlinear solve, used for initialization; no E3SM component has, to our knowledge, a discrete adjoint of its transient time stepper. RDycore is a finite-volume shallow-water dynamical core (Roe fluxes, well balancing, wetting and drying) validated at up to 471M cells. We add the missing derivative infrastructure — an exact assembled Jacobian and TSAdjoint-based sensitivities [19] — entirely behind a configuration flag, with no impact on existing runs and no new dependencies.

Contributions: (i) an exact, FD-verified Jacobian of the SWE right-hand side, including Roe dissipation, entropy-fix branches, wet/dry branches, and reflecting/Dirichlet/critical-outflow boundary conditions; (ii) adjoint gradients dJ/du_0 and dJ/dn with respect to water-height observations at gauge locations at multiple observation times, FD-validated to 10^{-5} or better; (iii) TAO-based recovery of per-region, per-cell and land-cover-class roughness fields, with a quantified study of when the observing network makes them identifiable at all; (iv) implicit (backward Euler) stepping and its adjoint, enabled by the same Jacobian, with an empirical demonstration that explicit friction fails at operational time steps on a Hurricane Harvey configuration; and (v) a catalog of the non-differentiable structures encountered — useful, we believe, to anyone retrofitting adjoints onto an

operational flood code.

2 Forward model

RDycore solves the 2D shallow water equations in conservative variables $u = (h, q_x, q_y)$, $\mathbf{q} = h\mathbf{v}$:

$$\partial_t h + \nabla \cdot \mathbf{q} = Q_r(x, t), \quad (1)$$

$$\partial_t \mathbf{q} + \nabla \cdot \left(\frac{\mathbf{q} \otimes \mathbf{q}}{h} + \frac{gh^2}{2} \mathbb{I} \right) = -gh \nabla z_b - gn^2 h^{-7/3} \mathbf{q} \|\mathbf{q}\|, \quad (2)$$

with cell-centered finite volumes on an unstructured mesh:

$$\frac{du_i}{dt} = -\frac{1}{|\Omega_i|} \sum_{e \in \partial\Omega_i} \mathcal{F}(u_i, u_j, \mathbf{n}_e) L_e + S(u_i; n_i, t) \equiv f_i(u, n), \quad (3)$$

where $\mathcal{F}$ is the Roe flux with a critical-flow (entropy) fix, primitives are reconstructed with a regularized division and a dry cutoff $h_{\min}$, and S contains bed slope, Manning drag, and external forcing.

One piece of vocabulary recurs throughout the paper. The flux and source terms are piecewise-defined: alternative closed-form expressions — *branches*, in the sense that $|x|$ has a positive and a negative branch — selected at run time by conditions on the state. The Roe flux’s dissipation is built from the moduli $|\lambda_p|$ of its characteristic wave speeds, so it switches branch wherever a wave speed changes sign, and the entropy fix substitutes a third, smoothed expression when a wave speed is near zero (a transcritical state); the dry cutoff switches drag and flux terms on and off at $h_{\min}$; boundary ghost maps switch with the flow direction. Away from a switching surface the map is smooth and equals its active branch, and the derivative machinery of Sections 3–4 is *branch-faithful*: it differentiates the expression the condition actually selected — the correct one-sided derivative.

A differentiable right-hand side. A prerequisite that deserves emphasis: RDycore’s existing friction treatments (a semi-implicit splitting and the implicit scheme of Xia and Liang) embed the time step — and, in the semi-implicit case, the flux divergence — *inside* the RHS callback. The resulting map is consistent only with forward Euler and admits no well-defined $\partial f / \partial u$. We therefore added a plain explicit source treatment (`source.method: explicit`), used on every adjoint path; Section 6 shows how the resulting stiffness is then handled properly, by the integrator rather than inside f .

3 Exact Jacobian

The RHS Jacobian is block sparse with 3×3 blocks on the cell-adjacency graph: each interior edge (i, j) contributes $\mp(L_e/|\Omega_{i,j}|) \partial \mathcal{F} / \partial u_{L,R}$ to the two cell rows, local sources contribute diagonal blocks, and boundary edges contribute diagonal blocks through the ghost-state map of the boundary condition.

Flux blocks by forward-mode differentiation. Rather than deriving the dissipation-matrix derivatives symbolically, we differentiate the flux *computation*: a hand-written forward-mode (tangent) version of the Roe flux routine propagates directionals through every intermediate — Roe averages, eigenvectors, eigenvalues including both branches of the critical-flow fix, wave strengths, and physical fluxes — taking the code’s own branch at every $|\cdot|$ and switch (a consistent one-sided

Check	Configuration	Result	Gate
Edge flux blocks	wet jump 10 m/5 m	7.6×10^{-9}	10^{-6}
Edge flux blocks	transcritical (entropy-fix branch)	2.4×10^{-10}	10^{-5}
Source block	shallow flowing state	$< 10^{-6}$	10^{-6}
Assembled matrix	uniform flow, reflecting BCs	1.6×10^{-8}	10^{-6}
Assembled matrix	dam break, reflecting BCs	1.5×10^{-8}	10^{-6}
Assembled matrix	dam break, Dirichlet + critical outflow	1.9×10^{-8}	10^{-6}
dJ/du_0 (explicit RK)	8 sampled components	1.0×10^{-8}	10^{-5}
dJ/dn (explicit RK)	domain aggregate	2.9×10^{-6}	10^{-5}
dJ/du_0 (implicit BE)	tight inner tolerances	1.4×10^{-8}	10^{-5}
dJ/dn (implicit BE)	tight inner tolerances	5.3×10^{-6}	10^{-5}

Table 1: Relative errors of analytic derivatives against central finite differences of the full nonlinear computation. Every check in this table is an automated test in RDycore’s `ctest` suite, run on every change to the repository.

derivative). Seeding unit directions through the primitive-reconstruction chain rule yields the two conservative-space blocks in six tangent evaluations per edge. The approach mirrors the flux code line by line, which makes it reviewable and, in our experience, correct on first assembly against the verification harness below.

Boundary blocks. For a boundary edge, $F(u_{\text{in}}, u_g(u_{\text{in}}))$ contributes $(\partial_L F + \partial_R F \partial u_g / \partial u_{\text{in}})$. The reflecting map is a linear mirror; Dirichlet ghosts are constant; critical outflow has a closed-form ghost map $h_g = (q^2/g)^{1/3}$ with the code’s inflow branch producing an identically zero flux.

Parameter Jacobian. Manning n appears only in the drag term, so $\partial S_{\mathbf{q}} / \partial n = -2gnh^{-7/3} \mathbf{q} \|\mathbf{q}\|$: two nonzeros per cell, supplied to `TSSetRHSJacobianP`.

Verification. Three layers, all against central finite differences, with the pass/fail tolerances fixed in advance of the runs (Table 1): an edge-level harness that differences the flux routine itself on sampled states (with a Richardson self-check of the FD order); full-matrix comparisons of the assembled Jacobian against a matrix-coloring FD Jacobian of the actual RHS on three state/boundary combinations, boundary rows included; and end-to-end gradient checks through the adjoint (next section). A finite-difference coloring Jacobian (`numerics.jacobian: fd`) remains available permanently as the verification baseline.

4 Adjoint sensitivities

For gauge observations y_k at times t_k with observation operator H (one row per gauge, selecting water height) and error variance σ^2 ,

$$J(n) = \frac{1}{2\sigma^2} \sum_k \|Hu(t_k) - y_k\|^2. \quad (4)$$

`TSAdjoint` [19] differentiates the discrete time stepper, so gradients are exact to roundoff for the computed trajectory — the property that makes the FD gates in Table 1 meaningful. Multi-time observations are handled by a segmented backward sweep: the adjoint is advanced between observation times with `TSAdjointSetSteps` and receives a jump $H^T \sigma^{-2} r_k$ at each t_k (Algorithm 1

in Section 5). One sweep yields both dJ/du_0 (state sensitivity) and dJ/dn (per-cell parameter gradient, via the two-nonzero parameter Jacobian).

Two integrator facts constrain the design. PETSc’s plain forward-Euler integrator implements no adjoint, so adjoint-path configurations use the Runge–Kutta family. And for implicit integrators the discrete-adjoint gradient is only as exact as the inner solves: the Manning gradient fails its gate at default inner tolerances and passes once they are tightened (Table 1, implicit rows). Loose inner tolerances also turned out to be a false economy on this problem: a tighter Krylov tolerance makes the implicit forward *faster*, because better linear solves let Newton converge in fewer iterations, and the production setting (Krylov 10^{-4} , nonlinear 10^{-5}) reproduces the fully converged answer to every printed digit for about a quarter more wall time than the loosest usable setting. Campaigns therefore run at converged inner tolerances, and the verified gradients *are* the production gradients. What the FD gates cannot certify at production scale is a different matter entirely — the objective’s own nonsmoothness over long windows — quantified in Section 7.2.

5 The calibration algorithm: strong-constraint 4D-Var

The complete calibration loop is summarized in Algorithms 1–3. It is a strong-constraint 4D-Var iteration: the model is enforced exactly (the only controls are the Manning field and, optionally, the initial state). Each objective evaluation performs one checkpointed forward sweep that records the gauge residuals r_k at the observation times, followed by one backward sweep of the discrete adjoint (Algorithm 2). The per-iteration cost is therefore one forward plus one adjoint solve, independent of the number of parameters.

The backward sweep carries two vectors with distinct roles. The *adjoint state* λ holds $\partial J_{\text{mis}}/\partial u(t)$ at the current point of the backward traversal. It is seeded with the terminal observation sensitivity $\sigma^{-2}H^T r_K$, propagated one step at a time by the transposed Jacobian of the forward step map (the backward “model step”), and incremented by a jump $\sigma^{-2}H^T r_k$ each time the sweep crosses an observation time, because the state at that time enters the misfit directly. The *parameter sensitivity* μ starts at zero and accumulates, at every step, the chain-rule contribution $(\partial \Phi_s/\partial n)^T \lambda$ of that step’s Manning dependence. When the sweep reaches $t = 0$, μ is the complete misfit gradient dJ_{mis}/dn and λ is dJ_{mis}/du_0 . Both transposed Jacobians are the analytic operators of Sections 3–4 — the assembled RHS Jacobian for the state propagation, the two-nonzero parameter Jacobian for the μ accumulation — applied within TSAdjoint’s discrete adjoint of the chosen integrator.

The outer update is TAO’s bound-constrained limited-memory quasi-Newton method BLMVM [20] (Algorithm 3). The loop terminates on the standard bound-constrained test: the projected gradient g^P — the gradient with components pointing out of the feasible box zeroed — drops below an absolute tolerance, or an iteration budget is reached. The same loop serves every parameterization in this paper: per-region, per-cell, and land-cover-class fields differ only in the expansion of the parameter vector onto cells (line 3 of Algorithm 1) and the corresponding summation of μ back onto parameters.

Practical execution. One objective evaluation costs one checkpointed forward sweep plus one adjoint sweep, independent of the number of parameters; each trial step of the BLMVM line search costs one such evaluation. A basin-scale window needs more iterations than a single batch allocation provides, so a run writes its parameter vector on exit and the next reads it back as its starting iterate, matched by land-cover class rather than by position. The resume is exact in the parameters — the continuing run’s first objective reproduces the previous run’s last to six digits — but BLMVM’s stored curvature pairs do not survive it, so a chain of jobs is not step-for-step

Algorithm 1 Strong-constraint 4D-Var calibration of the Manning field, as implemented with TAO/BLMVM and TSAdjoint. The window map $M_{t_{k-1} \rightarrow t_k}$ is the composition of m steps of the discrete forward stepper (explicit RK, backward Euler, or ARK-IMEX); Algorithm 2 differentiates exactly this stepper, so the gradient is exact to roundoff (and inner-solver tolerance, Section 4) for the computed trajectory.

Require: prior/initial guess n_0 , bounds $[n_{\min}, n_{\max}]$, observations y_k at times $t_k = km \Delta t$ ($k = 1, \dots, K$), gauge operator H , error σ , Tikhonov weight β , gradient tolerance ε_g , iteration budget $I_{\max}$

- 1: $n \leftarrow n_0$
- 2: **repeat**
- 3: set the model’s cellwise drag coefficient from $n \triangleright$ identity for per-cell n ; constant over each region or land-cover class for the reduced parameterizations
- 4: $u \leftarrow u_0$ $\triangleright$ fixed initial state (strong constraint)
- 5: **for** $k = 1, \dots, K$ **do** $\triangleright$ forward sweep; trajectory checkpointed — every step, or a revolve schedule for long windows (Section 6)
- 6: $u \leftarrow M_{t_{k-1} \rightarrow t_k}(u; n)$
- 7: $r_k \leftarrow Hu - y_k$
- 8: **end for**
- 9: $J \leftarrow \frac{1}{2\sigma^2} \sum_{k=1}^K \|r_k\|^2 + \frac{\beta}{2} \|n - n_0\|^2$
- 10: $(\lambda, \mu) \leftarrow \text{ADJOINTSWEEP}(\{r_k\})$ $\triangleright$ Algorithm 2: $\mu = dJ_{\text{mis}}/dn$
- 11: $g \leftarrow \mu + \beta(n - n_0)$
- 12: $n \leftarrow \text{BLMVMSTEP}(n, J, g)$ $\triangleright$ Algorithm 3
- 13: **until** $\|g^P\| \leq \varepsilon_g$ or $I_{\max}$ iterations
- 14: **return** n

Algorithm 2 ADJOINTSWEEP: segmented discrete-adjoint backward sweep (TSAdjointSolve with TSAdjointSetSteps). Φ_s denotes the forward map of step s , $u_s = \Phi_s(u_{s-1}; n)$; its transposed Jacobians are built from the assembled RHS Jacobian $\partial f/\partial u$ (Section 3) and the parameter Jacobian $\partial f/\partial n$ according to the integrator’s discrete-adjoint formula.

Require: checkpointed trajectory $\{u_s\}_{s=0}^{Km}$, residuals $\{r_k\}_{k=1}^K$ at steps $s = km$, gauge operator H , error σ

Ensure: $\lambda = dJ_{\text{mis}}/du_0$, $\mu = dJ_{\text{mis}}/dn$

- 1: $\lambda \leftarrow \sigma^{-2} H^T r_K$ $\triangleright \lambda$ holds $\partial J_{\text{mis}}/\partial u(t_s)$ throughout
- 2: $\mu \leftarrow 0$
- 3: **for** $s = Km, \dots, 1$ **do**
- 4: restore u_{s-1} (and stage values): read the checkpoint, or recompute forward from the nearest one when the checkpoint schedule demands
- 5: $\mu \leftarrow \mu + (\partial \Phi_s/\partial n)^T \lambda$ $\triangleright$ accumulate this step’s Manning dependence
- 6: $\lambda \leftarrow (\partial \Phi_s/\partial u_{s-1})^T \lambda$ $\triangleright$ backward model step (transposed step Jacobian)
- 7: **if** $s - 1 = km$ for some $1 \leq k < K$ **then**
- 8: $\lambda \leftarrow \lambda + \sigma^{-2} H^T r_k$ $\triangleright$ observation jump: u_{km} enters J_{mis} directly
- 9: **end if**
- 10: **end for**
- 11: **return** (λ, μ)

Algorithm 3 BLMVMSTEP: one iteration of TAO’s bound-constrained limited-memory variable-metric method [20]. $P_{[n_{\min}, n_{\max}]}$ is componentwise clipping to the bound box; the curvature pairs $\{(s_i, y_i)\}$ persist across calls. The projected gradient used in the convergence test of Algorithm 1 is $g_i^P = g_i$ for n_i strictly inside the box, $\min(g_i, 0)$ at the lower bound, and $\max(g_i, 0)$ at the upper bound.

Require: iterate n , objective J , gradient g , stored pairs $\{(s_i, y_i)\}$, bounds $[n_{\min}, n_{\max}]$

- 1: $d \leftarrow -B g^P$ $\triangleright B \approx (\nabla^2 J)^{-1}$ by the L-BFGS two-loop recursion over the stored pairs
- 2: find $\alpha > 0$ with a projected line search on $n(\alpha) = P_{[n_{\min}, n_{\max}]}(n + \alpha d)$ satisfying sufficient decrease of J $\triangleright$ each trial α costs one forward-plus-adjoint evaluation (lines 3–11 of Algorithm 1)
- 3: $n^+ \leftarrow n(\alpha)$; evaluate g^+ at n^+
- 4: $s \leftarrow n^+ - n$; $y \leftarrow g^+ - g$; store (s, y) if $s^T y > 0$, discarding the oldest pair beyond the memory limit
- 5: **return** n^+

identical to one long run. On a twin at the Houston mesh the restart was not a penalty: a chained 2+2 iteration run reached a lower objective than a single cold four-iteration run, the restart acting as a rescaling of the quasi-Newton model. That is one twin, not a rule; the point for planning is that chaining did not cost convergence there.

Scaling the optimization problem. Algorithm 1 is written in physical Manning units, and at basin scale that is not a usable parameterization. The quasi-Newton step of Algorithm 3 begins from $B = I$, so the first trial point is $n - g$; on the production window $\|g\| \approx 1.5 \times 10^3$ while $n \approx 0.05$, five orders apart. The projection onto $[n_{\min}, n_{\max}]$ then sends every component to the lower bound — which is exactly where the model cannot solve, since a uniform $\alpha = 0.2$ diverges (Table 7) — so every line-search trial fails and the optimizer terminates having taken no step at all. Nothing is wrong with the gradient when this happens; the finite-difference gates pass throughout.

Two changes make the problem well scaled, and both have physical readings rather than being tuning constants. First, optimize the *relative* field $\alpha_k = n_k/n_{\text{prior},k}$ and the relative objective $J/J(n_{\text{prior}})$. Both are then $O(1)$ and the unit quasi-Newton step becomes a roughness change of order 10%: at the NLCD start point of the production configuration, over a shortened window that reproduces the zero-step failure exactly, $\|g\|/J_0 = 0.10$, and the same start point that took no step in physical units takes one that lowers the objective and the peak-WSE MAE together. The bounds become statements about relative roughness, $\alpha \in [0.3, 3]$, which additionally excludes the region where the implicit solve fails. Second, set the Tikhonov weight from a prior standard deviation on Manning n rather than by hand,

$$J = \frac{1}{2\sigma^2} \sum_k \|r_k\|^2 + \frac{1}{2\sigma_n^2} \|n - n_{\text{prior}}\|^2, \quad \beta = \sigma_n^{-2}, \quad (5)$$

and report σ_n . This matters because β is dimensional: a bare value is uninterpretable across problems, and ours had been chosen independently in every experiment, spanning six orders of magnitude. Against the measured $J(\alpha)$ curve, $\sigma_n = 0.015$ places the minimum of the dominant identifiable direction at $\alpha \approx 0.70$, while $\sigma_n = 0.020$ leaves no interior minimum there at all — the same pathology the hand-chosen weight had. Stating the prior turns the choice into an arithmetic consequence of a quantity the domain team can argue about.

The diagnosis is worth separating from the fix, because the failure mode is not specific to this model: an optimizer that takes zero steps on a problem whose gradient passes every verification

gate is a scaling defect, and the cheapest test is to compare $\|g\|$ against the scale of the variable before suspecting the derivative.

6 Implicit stepping, enabled and required

The explicit drag term is stiff in thin water: linearizing gives a stability limit $\Delta t \lesssim 2h^{4/3}/(gn^2|v|)$, which for sheet flow at centimeter depths is seconds — far below the gravity-wave CFL limit of kilometer-scale meshes. Our Hurricane Harvey test configuration (Houston at 1 km, $\Delta t = 30$ s) confirms this empirically: the explicit source treatment produces NaN within the first objective evaluation, in quantitative agreement with the analysis. The same holds at basin scale, where the margin is far wider: on the spun-up 30 m, 2.93M-cell Harvey mesh (99.9% wet, median depth 6 mm) the friction-rate limit is $\Delta t \lesssim 2.4$ ms, and forward Euler of the differentiable right-hand side returns a non-finite state at every tested step ($\Delta t = 0.25, 0.125, 0.05$ s) while backward Euler at $\Delta t = 1$ s completes a 72-hour window without a single Newton failure. There is no viable explicit path for the differentiable right-hand side at this resolution.

The assembled exact Jacobian makes the remedy nearly free: backward Euler via PETSc’s theta method, with SNES driven by the same Jacobian, carries the Harvey window at $\Delta t = 30$ s, and the theta-method adjoint passes the same FD gates (Table 1).

We also implemented the sharper tool: an ARK-IMEX splitting with the Manning drag as the stiff implicit part (`temporal: arkimex`). The implicit stage system is block-diagonal — an independent 3×3 Newton per cell, no global solve — while fluxes and bed slope remain explicit under the gravity-wave CFL limit. Its adjoint passes the same gates (2.0×10^{-8} for dJ/du_0 , 2.5×10^{-6} for dJ/dn) and it carries the Harvey window at $\Delta t = 30$ s at 6.7 s per forward-plus-adjoint sweep, with gradients matching backward Euler to three digits. The splitting also runs under RDycore’s libCEED operator backend — the explicit side uses the friction-free source Q-functions prepared in PR #359, and the two backends agree to machine precision (5×10^{-16} relative) after 100 ARK-IMEX steps; a device-resident implicit part is the natural follow-up.

Three implementation facts others will hit:

- PETSc’s ARKIMEX adjoint dereferences *both* the implicit- and explicit-part parameter Jacobians, so a zero explicit-part `TSSetRHSJacobianP` must be registered even when the parameter appears only implicitly.
- The embedded step-size controller must be disabled for finite-difference gradient validation: perturbed solves otherwise take different step sequences, which shows up as an apparent $\sim 10^{-4}$ gradient error that vanishes with fixed steps.
- PETSc’s in-memory checkpointing trajectory cannot tolerate a *segmented* forward solve (one `TSSolve` per observation window). `TSSolve` sets its converged reason before the final trajectory write of every call, which the memory trajectory interprets as the end of the forward run, corrupting the checkpoint schedule at every interior observation time. The forward sweep must be a single `TSSolve` with observations recorded by a monitor; the segmented backward sweep is unaffected.

Newton robustness is set by residual discontinuities, not by conditioning. Where implicit stepping fails on this problem, it fails for a reason that is invisible in the smooth twin experiments: the residual is discontinuous, so Newton has nothing to converge *to*. At the land-cover roughness prior, rather than the twins’ smooth low- n fields, backward Euler descends cleanly for a

Rain-forced configuration (land-cover prior)	Outlet	Outcome
600-step control	critical outflow	Newton pinned from step 498
600-step control	free outflow	zero failures; 598/600 solves in two iterations
1-h calibration window (forward + adjoint + line search)	free outflow	clean end to end
72-h forward through the Harvey crest (2.93M cells)	free outflow	259,200 solves, zero failures

Table 2: The outlet replacement is a cure, not an improvement. Under critical outflow, tolerance relief bought steps only linearly; the transmissive outlet completes the identical window and every longer one attempted.

dozen iterations and then freezes — a vanishingly small line-search step changes the residual norm severalfold, the signature of a jump, not of ill-conditioning. Two mechanisms are responsible, both catalogued in Section 6.1 and both invisible until the parameter field is realistic: the drag’s wet/dry cutoff, whose amplitude scales as n^2 , and the critical-outflow boundary’s stagnation switch.

Both fixes are implemented, gated, and opt-in (open issues 4–5). Regularizing the drag velocity removes the first and restores honest convergence over the early transient. The second we removed by replacing the outlet condition itself: a transmissive (zero-gradient) free outflow — the ghost state copies the interior state, the convention of most finite-volume shallow-water codes — is continuous in the interior state by construction, and its ghost map is the identity, so the analytic Jacobian and adjoint carry it exactly, FD-gated like every other block. The effect is not an improvement but a cure (Table 2): the identical failing control completes with zero nonlinear failures, and the fix holds through the longest windows we run. Before the replacement we absorbed the pin by loosening the nonlinear tolerance; a class-twin calibration under that crutch still recovered all fifteen class values to machine precision, so the gradient path was unimpaired — tolerable, but a crutch. The general lesson for adjoint retrofits is that nonsmoothness in an operational flood code is not only a differentiability question — it is a nonlinear-solver question.

Checkpointed adjoints for long windows. A naive stored trajectory for a full 36-hour Hurricane Harvey window at the 30 m mesh’s CFL-limited $\Delta t = 0.25$ s (518,400 steps, 2.93M cells) would occupy tens of petabytes. PETSc’s checkpointing trajectory (`-ts_trajectory_type memory` with the revolve optimal-schedule library) replaces storage with recomputation. We validated it across the full hierarchy. On the dam-break twin it reproduces the disk-trajectory calibration and gradient gates bit-for-bit for explicit RK, backward Euler, and ARK-IMEX (the implicit integrators need the same tight inner tolerances as in Section 4, since checkpoint recomputation re-solves the stage systems). On the 30 m mesh, a forward-plus-adjoint sweep with only *three* states checkpointed in memory matches the disk-trajectory gradient to all printed digits — and runs slightly faster, because recomputation is cheaper than trajectory I/O at this scale. For the full window, a few hundred checkpoints (~ 14 GB aggregate) bound the recompute overhead by a factor of about four, making the long-window adjoint memory-feasible; the remaining cost is the forward compute itself, addressed next.

6.1 Nondifferentiability catalog

[Reviewers: do we want this subsection at all? It is the least conventional part of the paper. Five of its six entries are now resolved with gated fixes, which is an argument for keeping it as the record of what retrofitting an adjoint onto an operational flood code actually costs — and the counter-argument is that a resolved catalog is engineering detail that belongs in an issue tracker.

Cut, keep here, or move to an appendix.]

Retrofitting an adjoint onto an operational flood code surfaces nonsmooth structure worth documenting. Every entry below is annotated with its status: five are resolved and gated in the test suite, and one remains open.

- **Time step inside the RHS.** (*Resolved.*) Operator-split friction treatments (here: semi-implicit and XQ2018) make f depend on Δt and on intermediate flux divergences; no consistent $\partial f / \partial u$ exists. The remedy is an explicit f plus a genuinely implicit integrator, not a smarter derivative — which is the architectural choice this section exists to justify, and the reason the implicit-versus-explicit cost comparison above is not the one a forward-only code would make.
- **Entropy-fix branches.** (*Resolved.*) The critical-flow fix is piecewise smooth; branch-faithful one-sided derivatives pass FD checks even on transcritical states (Table 1).
- **Critical-outflow stagnation.** (*Resolved by changing the boundary condition.*) The outflow ghost map $h_g = (q^2/g)^{1/3}$ has an unbounded one-sided derivative at $q = 0$, and the inflow branch switches there: a genuine subgradient point. Unlike the others in this list it is not merely a derivative subtlety — the branch switch is a discontinuity of the *residual* itself, since the $u_\perp < 0$ branch zeroes both states (a wall) while the $u_\perp \geq 0$ branch imposes the critical ghost, so the edge flux jumps by the full wet-onto-dry Roe flux, $O(gh^2/2)$, as a boundary cell crosses stagnation. With an implicit integrator this stalls Newton outright: the line search collapses to $\lambda \sim 10^{-6}$ with the residual pinned at a floor of 10^{-4} – 10^{-5} relative. Explicit stepping never sees it. The cure is to change the boundary condition, not the solver: a transmissive (zero-gradient) free outflow — ghost state equal to the interior state, the standard free outflow of finite-volume shallow-water codes — is continuous by construction, has an identity ghost map in the Jacobian, and removes the stall entirely. Provided as an opt-in condition alongside the original, and gated: a 72-hour rain-forced basin forward completed 259,200 implicit solves with zero nonlinear failures.
- **Drag at the wet/dry cutoff.** (*Resolved by regularizing the drag velocity.*) The drag $gn^2h^{-7/3}\mathbf{q}\|\mathbf{q}\|$ gated at $h_{\min}$ with a plain $\mathbf{q}/h$ velocity is likewise residual-discontinuous: a cell crossing the cutoff with retained momentum switches an $O(h^{-7/3})$ term on and off. The amplitude scales with n^2 , so it is invisible in smooth- n twin experiments and disabling at land-cover roughness. Evaluating the drag with the ANUGA-regularized velocity $\mathbf{u} = \mathbf{q}h/(h^2 + h_{\text{anuga}}^2)$ removes it (the drag then vanishes as $h^{5/3}$), and reduces identically to the plain form at $h_{\text{anuga}} = 0$. The value of h_{anuga} itself is an empirical choice rather than a physical one (open issue 5).
- **Wet/dry branches.** (*Resolved in the derivative; measured as objective nonsmoothness at basin scale.*) That a moving wet/dry front is the awkward part of a shallow-water adjoint is well known: Funke et al. built an adjoint wetting and drying scheme precisely to make derivative-based optimization usable for inundation problems [21]. In our setting the RHS's dry cutoffs are honored branch-faithfully in every block; fully wet validation cases keep FD comparisons clean, and dry cells are correctly insensitive to their roughness. At basin scale the branches also make the *objective* nonsmooth: over long rain-forced windows the trajectory crosses so many per-cell depth-cutoff surfaces that a finite-step central difference measures a smoothed slope rather than the one-branch derivative the adjoint returns — quantified, with the verification policy it implies, in Section 7.2.

- **The argmax in the peak-WSE observable.** (*Open — the one entry we have not closed.*) The high-water-mark misfit (Section 7.2) depends on the state through $\max_k Hu(t_k)$: piecewise smooth, with kinks where two sampled times tie for a mark’s peak. The adjoint injects each mark’s residual at its recorded argmax time, which is the exact gradient away from ties (a measure-zero set); but the peak time *moves* across optimizer iterates, so the objective the quasi-Newton method sees is only piecewise smooth and its line search can straddle kinks. Practice is untroubled — the FD gates pass away from ties and the twin calibrations descend monotonically — and the instrumented basin-scale gate adds a measurement: across every FD probe on the real 108-mark configuration, *no* mark’s argmax moved and none flipped wet/dry, so at this geometry the argmax kinks are far sparser than the dynamics’ wet/dry kinks (previous entry). A second, independent measurement points the same way: evaluating the production window on 4, 8 and 16 MPI ranks — three different domain decompositions, hence three different reduction orders — reproduces the peak-WSE objective and MAE to every printed digit ($J = 8.082566 \times 10^2$, MAE 0.7188 m), so no mark’s recorded peak time is sensitive to partitioning at the level roundoff perturbs it. This remains the standard footing of max-type observables in calibration rather than a theorem. *Proposed remedy, not yet implemented:* freeze the argmax assignment across an outer iteration, so each inner subproblem is genuinely smooth in the parameters, and re-evaluate the peak times between outer iterations — a fixed point on peak times, converged when the assignment stops changing. This costs one extra forward per outer iteration and makes the smoothness assumption the line search relies on true by construction rather than true in practice. A smooth surrogate (softmax over the sampled times, tempered toward the true maximum) is the alternative, at the cost of biasing the peak it calibrates against.

7 Hurricane Harvey twin experiments: regularization and the real-network limit

The per-cell twin runs on the Houston 1 km Hurricane Harvey mesh (2,746 cells, 4200 s window at $\Delta t = 30$ s). Truth is a two-zone field split at the domain midline, observed at 343 synthetic gauges over 14 observation times, calibrated with implicit stepping and Tikhonov regularization ($\beta = 10^{-5}$). Every cell is wet and moving in this configuration, so all 2,746 parameters are observable. The configuration is deliberately gauge-sparse — 343 gauges for 2,746 parameters, an 8:1 ratio — and it exhibits the classic *semiconvergence* of underdetermined inverse problems (Figure 1). At 300 BLMVM iterations (~ 22 minutes on one laptop core, ~ 4.4 s per forward-plus-adjoint) the recovered map shows the two-zone truth structure cleanly at 26.3% relative L^2 error. Continuing the same weakly regularized optimization to 2000 iterations *degrades* recovery to 38.0%: the misfit keeps decreasing while the parameters drift along observation-null directions, ultimately pinning cells against both bound constraints (Figure 1, center left). Early stopping was acting as the effective regularizer. The behavior disappears when the problem is made well posed: a properly regularized Houston run ($\beta = 10^{-4}$, 686 gauges) reaches 19.4% at a 1000-iteration budget with no drift (Figure 1, center right) — the sharpest recovery of the three, with the zone boundary resolved at the truth midline. This is precisely the overfitting risk documented for distributed friction estimation [8], here reproduced in a controlled twin setting. It motivates the regularization studies (total variation, land-cover priors, discrepancy-principle stopping) planned as follow-on work.

The real gauge network. The synthetic gauge sets above are a controlled ramp; the network that matters is the real one. Twenty USGS gauges fall inside the Houston mesh, seventeen with

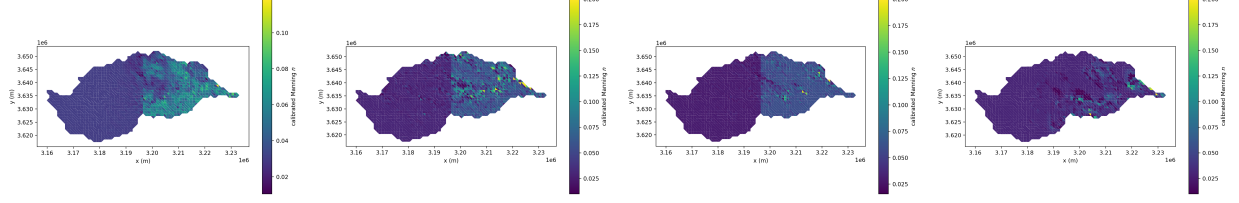


Figure 1: Regularization, semiconvergence, and network density on the Houston twin (2,746 parameters; truth: $n = 0.03$ west, $n = 0.06$ east of the midline; implicit stepping, 14 observation times). Far left: weak regularization ($\beta = 10^{-5}$, 343 gauges) stopped early at 300 iterations — 26.3% error. Center left: the same setup run to 2000 iterations — the misfit is lower but the field drifts along unobserved directions to 38.0% error and saturates the bounds $[0.01, 0.2]$. Center right: proper Tikhonov weight ($\beta = 10^{-4}$, 686 gauges), 1000 iterations — 19.4% error, no drift, sharp zone boundary. Far right: the *real* network — the 17 in-mesh USGS gauges with Harvey stage records, same $\beta = 10^{-4}$, 1000 iterations — the misfit falls 2×10^7 -fold while the field recovers to only 44.9%: no trace of the two-zone truth, near-prior almost everywhere with null-space structure pinned against both bounds.

Hurricane Harvey stage records (HydroShare DOI 10.4211/hs.c037167e497546a1bc1508dfb32a9cff). Re-running the same twin with the observation operator built at those seventeen gauge cells — 238 observations (17 gauges $\times$ 14 times) against 2,746 parameters — is a sobering experiment. BLMVM drives the misfit down by a factor of 2×10^7 , an essentially perfect fit of the observations, while the recovered field is 44.9% from the truth and saturates both bounds (Figure 1, far right). This is semiconvergence in its purest form: the real network’s null space is so large that data fit says almost nothing about parameter recovery, and no Tikhonov weight or stopping rule rescues a per-cell field from 238 observations. The practical conclusion for real-data calibration is to shrink the parameter space to match the network — but, as the next subsection measures, shrinking the parameter space is necessary and not sufficient: whether a reduced parameterization is identifiable depends on the observation geometry and window, not on the ratio of observation to parameter counts.

7.1 Identifiability is a property of the observable: gauges versus high-water marks at 30 m

The natural reduced parameterization for this domain is one Manning value per NLCD land-cover class (15 classes at the 30 m Turning mesh, 2.93M cells), with the published NLCD lookup as prior and truth for twins. Counting is deceptively favourable: even the sparse real network supplies an order of magnitude more observations than parameters. Measurement says otherwise. In a class twin on the 30 m mesh at the real 13-gauge geometry (260 observations, a short early-transient window), the objective falls $128\times$ while the recovered class values remain 66% wrong in relative L^2 , worst class 81% — and the failure is structural: classes whose cells are dynamically distant from every gauge (the three forest classes) never move from the starting value, because within the window nothing their roughness does can reach a gauge. The identical configuration observed at 418,076 strided cells recovers all fifteen classes to machine precision, so the failure is the network, not the method. Fifteen parameters, thirteen gauges, and an “overdetermined” observation count: still unidentifiable.

The same question re-asked with the peak observable comes out differently. Observing the

same NLCD-truth twin at the 108 QC-passed high-water-mark cells (peak WSE sampled over a one-hour rain-forced window, implicit stepping, free-outflow outlet; Section 7.2) and calibrating fifteen classes from a uniform start: the objective falls $81\times$, the peak-WSE error at the marks drops from 0.103m to 0.017m MAE, and the area-dominant classes are identified — the three developed-intensity classes, pasture, and crops recover to 0.7–10.5%, together 83% of the domain’s cells; area-weighted relative L^2 error is 0.20 (unweighted 0.58). The residual errors concentrate in small-area classes (forests, shrub, emergent wetland, $\sim 5\%$ of cells), which now *move* — they are sensitive, unlike in the gauge experiment — but overshoot to the bounds along weakly constrained directions: ordinary semiconvergence, treatable with regularization and longer windows rather than a structural absence of signal. Three caveats keep this honest: the one-hour window samples early-transient peaks, not flood peaks (the real calibration windows are 6–12 hours and should only help); the run was capped at 15 quasi-Newton iterations; and twin marks synthesized from the truth trajectory carry no representation error, so this is an identifiability statement — an upper bound on real-data performance, not a forecast of it.

Noise turns identifiability into a signal-to-noise question. Every recovery above is noise-free, as are the verification twins behind Section 5. Such experiments establish that the gradient is right and that the observable carries the information; they do not establish that a field can be calibrated from real data. What separates the two is the size of the parameter-induced signal relative to observation error, and that ratio moves by orders of magnitude with the configuration (Table 3). On a short verification window the entire Manning-induced observation signal is a few microns of water height, so a noise-free optimizer descends on it indefinitely while a tenth of a millimetre of observation noise destroys the recovery outright — and no Tikhonov weight in a sweep up to $\beta = 1$ rescues it. At basin scale the same quantity is ~ 0.1 m of peak water-surface elevation after one simulated hour, five orders of magnitude larger, which is the regime in which survey-grade high-water-mark error becomes a fair fight rather than a foregone conclusion. The lesson is not that the method is fragile: it is that friction must act long enough to imprint a signal above the noise floor, which makes window length a first-class calibration decision rather than a cost parameter. The basin-scale noisy row shows what marginal signal-to-noise looks like in practice, and why misfit at the marks is the objective but not the result. With 0.15 m of observation noise, even the *true* field cannot beat a mark MAE of about $\sigma\sqrt{2/\pi} \approx 0.12$ m against the noisy values — and the optimizer reaches exactly that floor while driving a third of the classes onto their bounds (pasture, 428k cells, to the ceiling) and ending far from the true field it was supposed to recover. The data fit is as good as truth could achieve; the parameters that achieve it are wrong, because the last of the misfit was removed by matching noise. On real data the parameter error is invisible and the bound-pinning is its symptom — which is the concrete argument for stronger regularization, a longer window, and discrepancy-principle stopping (stop at the noise level, do not descend past it) in the production run.

7.2 Changing the observable: high-water marks

Calibration against the actual stage records is blocked by geometry, not software: at all 13 rain-driven USGS gauges in the 30 m domain, modelled water-surface elevation sits 1–11 m *above* observed stage, because a 30 m cell stores one mean ground elevation — essentially the bank of an incised bayou — while the gauge hangs in the channel below it; the offset is time-varying as the real channel fills, so no per-gauge datum correction can repair it. We therefore changed the observable to the USGS high-water-mark archive: surveyed *peak* water-surface elevations on structures and debris lines, i.e. on the floodplain surfaces a 30 m cell does represent, and the benchmark currency of

Configuration	Signal	Observation noise	Outcome
Verification twin (planar, 40 steps, 1.8 s)	5 μm water height	none 0.1 mm 1 mm	per-region recovery 4×10^{-7} per-cell recovery destroyed ($0.20 \rightarrow 1.8$) per-region destroyed ($4 \times 10^{-7} \rightarrow 0.83$, both at bounds)
30 m basin, 1 h rain-forced, 108 marks	~ 0.1 m peak WSE	none 0.15 m (HWM grade)	area-weighted rel. L^2 0.20; MAE $0.103 \rightarrow 0.017$ m misfit falls $9.4\times$, MAE $0.23 \rightarrow 0.12$ m — but 5 of 15 classes p

Table 3: Parameter-induced observation signal versus observation error. The noise-free recoveries differ by seven orders of magnitude in the signal they descend on; only the basin-scale row is a fair test against survey-grade error (0.1–0.3 m for high-water marks). Noise is a deterministic, decomposition-independent perturbation of the twin observations, so the comparison is reproducible across rank counts.

the flood-modeling literature (Inunda reports 0.67 m MAE against Harvey high-water marks [10]). Of 2,364 Harvey marks, 324 fall in the Turning domain; applying to the marks the same QC that disqualified the gauges — is the surveyed elevation above the containing cell’s bed? — rejects 62% of them (they are channel surveys), leaving 108 marks of quality fair or better. The misfit compares each mark cell’s peak simulated WSE over the window, $\max_k Hu(t_k)$, against the surveyed value; its adjoint needs no new machinery, because each mark’s residual can be injected at its own argmax time in the segmented backward sweep of Algorithm 2, which is the exact gradient wherever the peak time is locally constant (Section 6.1). The parameter gradient of the peak misfit passes the same central-difference gates as every other block on the smooth verification twin. At basin scale the naive check fails, and a set of instrumented measurements pins down why, in a way that validates the gradient rather than the check (Table 4). No mark’s argmax moves and none flips wet/dry across any probe, so the discrepancy lives below the observable; the evaluations are deterministic to nine digits and the FD value is stable across a decade of probe step, which excludes noise, curvature truncation, and sparse kinks. Decisively, the gap *accumulates with trajectory length*: at the shortest window the gate passes at the inner-solve floor with every piece of machinery engaged, and it grows three orders of magnitude by basin window lengths. A proportional defect in $\partial f/\partial n$ or the adjoint accumulation would err the same fraction at any window length; trajectory-accumulated branch crossings — the probe shifts which cells cross the depth cutoff at some step — predict exactly this profile. Two further checks close it: as the probe shrinks, the FD converges to the adjoint direction by direction — the largest single-class direction passes the 10^{-5} gate outright at the smallest probe — and the fifteen per-class gaps *sum* to the domain-wide gap, the additivity that independent per-cell branch crossings predict and no plausible defect would arrange. The measured conclusion: the adjoint returns the exact one-branch derivative; over long windows the objective is dense with wet/dry branch surfaces, and a finite-step central difference converges to a locally *smoothed* slope that sits several percent from any one branch. The quasi-Newton iteration consumes one-branch derivatives of a piecewise-smooth objective — standard practice for max-type and switching systems — and its observed descent (Section 7.1) is consistent with that footing (Section 6.1).

The remaining scientific choice — confirmed or corrected by the domain team — is the calibration window, which for marks that record the event maximum must cover the flood peak.

The uncalibrated baseline against the real survey. Because the marks record the event maximum, the honest baseline evaluation must simulate through the crest: a 72-hour rain-forced forward from the available initial state (2017-08-26 18:00, reaching past the Buffalo Bayou crest), NLCD-prior Manning, free outflow, backward Euler at $\Delta t = 1$ s on the 2.93M-cell mesh — 259,200

Window	Steps	FD-vs-adjoint gap, domain-wide direction
smooth verification twin	40	9×10^{-7} (gate 10^{-5} : pass)
basin, 1 min	60	9×10^{-5} (inner-solve floor: pass)
basin, 5 min	300	8×10^{-3}
basin, 15 min	900	2×10^{-2}
basin, 30 min	1800	2×10^{-1}
basin, 1 h	3600	7×10^{-2}

Table 4: The FD gate’s failure accumulates with trajectory length on identical basin, rain, marks, and converged tolerances — the signature of branch crossings along the trajectory, not of gradient error. The 30-minute and 1-hour rows sample peaks over different horizons, so strict monotonicity across them is not expected; single-class directions sit one to three orders lower at every window.

Marks	n	MAE	bias
crest inside the 72-h window	71	1.51 m	+1.20 m
never crest (reach cannot drain)	37	7.06 m	+7.06 m
all	108	3.41 m	+3.21 m

Table 5: The uncalibrated baseline against the 108 surveyed high-water marks, decomposed. The second population is model-high at every single mark and identifies a reach with no working exit; the first is the calibration target.

implicit steps, *zero* Newton failures, two hours on one GPU node. The model wets all 108 marks, and the uncalibrated peak-WSE error decomposes into two distinct populations (Table 5; confirmed mark for mark at fully converged inner tolerances). The marks that crest inside the window carry the error a Manning field can plausibly address. The others never crest at all: their cells fill monotonically through the entire window — every one peaks at the final hour with zero recession, still rising while the first population is already draining — accumulating some ten metres of standing water whose late-window inflow arrives laterally, after the rain has ended. Those cells sit in the downstream Buffalo Bayou reach: the model cannot drain it, a conveyance/outlet defect that no roughness value can absorb and that would corrupt any calibration asked to fit it. The calibration target is therefore the 71 admissible marks; the drainage defect is reported as a finding in its own right; and the remaining bundle (initial-condition, rainfall, mesh-elevation, and datum error) is what the calibration must be prevented from absorbing along with the roughness signal. The literature’s calibrated benchmark on this event is 0.67 m MAE [10].

The defect is a gradient, not a switch. Sorting the marks by crest time splits them into three groups, and those groups turn out to be three *spatial bands* whose bias increases monotonically downstream (Table 6). The never-crest population is the end of a continuum, not a separate pathology: the intermediate band already carries +2.85 m of the same model-high signature. Two consequences follow. First, a calibration window has to be placed on the upstream band, where the model is physically sound — crests there fall in event hours 29–41, and the 46 marks in that band are the target of every production number below. Second, the intermediate band is *not* clean held-out data, which rules out the obvious validation design; out-of-sample skill has to be tested by cross-validation within the upstream band instead.

Group	n	mean lon.	bias	MAE
crest h29–41 (upstream)	46	−95.695	+0.30 m	0.72 m
crest h48–72 (middle)	25	−95.568	+2.85 m	2.96 m
never crest (downstream)	37	−95.440	+7.06 m	7.06 m

Table 6: The crest-timing groups are spatial bands, and the model-high bias grows monotonically downstream toward the reach with no working exit. The upstream band is the calibration target; the middle band is contaminated by the same defect at intermediate strength and cannot serve as held-out data.

α	n range	J	MAE	dry
0.20	0.005–0.032	implicit solve diverges		
0.30	0.008–0.048	6.737×10^2	0.6392	1/46
0.45	0.012–0.072	7.038×10^2	0.6614	1/46
0.60	0.016–0.096	7.406×10^2	0.6776	1/46
1.00 (NLCD)	0.027–0.160	8.083×10^2	0.7188	0/46

Table 7: Uniform roughness scan on the production configuration (12-hour window, 46 cluster-A marks). The entire uniform-roughness knob is worth 0.08 m against a 0.72 m error, and only at roughness 30% of published values. The limit at $\alpha = 0.2$ is numerical, not drying — one mark stays dry throughout.

7.3 How much of the error can roughness explain?

Before asking an optimizer to fit fifteen land-cover classes it is worth asking what any roughness field could achieve, and that question has an optimizer-independent answer. Scaling the entire NLCD field by a single factor α traverses the most identifiable direction of the problem; evaluating the peak-WSE misfit along it bounds the authority of roughness over this observable without reference to how, or whether, a calibration converges.

We ran that scan on the production configuration: a 12-hour window over the upstream crest cluster (event hours 29–41), initialized from an hourly checkpoint of the 72-hour forward and with the rain re-aligned to the window start (both validated bit-exactly), scored against the 46 surveyed marks whose modelled crest falls inside the window. Each point is one forward with no adjoint (Table 7).

The entire uniform-roughness knob is worth 0.08 m against a 0.72 m error — about 11% — and it reaches that only at $\alpha = 0.3$, i.e. Manning values 30% of the published NLCD lookup, as low as $n = 0.008$ where smooth concrete is ≈ 0.012 . Over a physically defensible range ($\alpha \gtrsim 0.7$) the knob is worth about three centimetres. The curve is mildly concave with no interior minimum: J falls monotonically toward lower roughness until the implicit solve fails.

That number is corroborated from an independent direction. The identifiability twin of Section 7.1 measured the peak-WSE signal induced by a large class perturbation at ~ 0.1 m — the same authority scale, from a different experiment. So 85–90% of the model-versus-survey residual is not roughness at all: it is rainfall forcing, mesh elevation, datum, and representation error at 30 m.

The observable constrains one mode, not fifteen. A natural objection is that spatial redistribution should beat uniform scaling — that fifteen classes have freedom a single α does not. We tested it directly by scaling only part of the domain to $\alpha = 0.45$ and leaving the rest at the NLCD

Field	J	ΔJ	MAE	ΔMAE
NLCD prior	808.3	—	0.7188	—
developed classes 21–24 only (71.6% of cells)	786.8	−21.5	0.7306	+0.0118
everything else (28.4%)	827.5	+19.2	0.7268	+0.0080
both (= uniform $\alpha = 0.45$)	703.8	−104.5	0.6614	−0.0574

Table 8: Each half of the domain scaled alone makes the fit *worse*; scaled together they help by $45\times$ the sum of the parts. Peaks are set by path-integrated conveyance along flow paths that cross both groups, so only domain-coherent changes drain them.

prior (Table 8).

Each half alone makes the fit *worse*, and the two together help by $45\times$ the sum of the parts — a strong positive interaction, not two competing signs. The mechanism is consistent with every other measurement: a mark’s peak is set by the *path-integrated* conveyance from upstream to that mark, and those paths cross both groups. Speeding part of a path does not drain it; the resulting fast/slow junction ponds water, so partial changes *raise* peaks. Only a domain-coherent change lowers them.

The conclusion is a design finding, not a tuning observation. The peak-WSE observable at these marks constrains essentially *one* roughness degree of freedom — the domain-scale, path-integrated conveyance — worth $\sim 10\%$ of the model–survey discrepancy. It does not carry fifteen classes’ worth of information. This unifies the earlier evidence rather than sitting beside it: the perfect-data twin recovers only the classes that move the domain mean, and the noisy twin pins the others to bounds precisely because they carry no signal to fit, so the optimizer fits noise in those directions instead.

Relation to equifinality. None of this is a new phenomenon: that many different parameter sets reproduce a hydrological observation about equally well is Beven’s equifinality thesis [22, 23], and the usual response is to characterize the acceptable set by sampling it. What an adjoint changes is the cost of the diagnosis. Where GLUE-style methods *sample* the equifinal set with many forward runs, the measurements above *measure* its dominant direction directly: four forward evaluations bound the authority of the whole roughness field, and two more show that the modes are not separable. The result is the same kind of statement, obtained at a cost that does not grow with the number of parameters — which is what makes it affordable before a calibration rather than after one.

Why this generalizes. The measurement uses nothing specific to roughness. Given a parameter, an observable, and a model error to explain, the same three steps — scan the dominant parameter direction, check whether sub-domain changes are separable, and compare the resulting authority against the residual — bound what calibrating that parameter can ever accomplish, before any optimizer is written. We suggest it as a routine precondition for calibration studies, which commonly report the misfit reduction achieved without establishing the ceiling it was approaching.

8 Conclusions and outlook

An exact, verified Jacobian turns out to be the single artifact from which everything else follows cheaply: adjoint state and parameter sensitivities, gradient-based calibration, and implicit stepping. The capability is production-ready in the sense that matters for an operational code — behind a

flag, FD-gated in CI, and with the verification baseline (FD coloring) kept alive in the configuration system.

The second artifact is a measurement rather than a capability, and it is the one we would most like others to copy. Asking what fraction of a model’s error a parameter is *able* to explain — here, at basin scale, $\sim 10\%$ at the edge of physical plausibility and $\sim 4\%$ within it (Section 7.3) — is cheap, optimizer-independent, and it reframes what a calibration result means. It also told us where the leverage actually is: the same machinery, pointed at the residual rather than at the roughness, isolated a reach the model cannot drain.

Next steps, in order: the converged real-data calibration — land-cover-class parameters against the marks that crest inside the window, with the scaled optimization problem of Section 5 and a stated σ_n , whose expected landing point the α -scan already predicts; calibrating the number of parameters the observable actually supports rather than the number the land-cover map offers, which our own evidence puts at one to three; the conveyance and outlet representation that the drainage finding implicates, to which the adjoint machinery applies unchanged; and coupling the gradient path to PyTorch through the existing PETSc-PyTorch bridge for learned-physics and hybrid data-assimilation experiments.

A closing note on cost. Checkpointing makes the long-window adjoint memory-feasible, but at the 30 m mesh’s CFL-limited $\Delta t = 0.25$ s a full 36-hour Hurricane Harvey gradient still costs several forward sweeps over half a million steps. The affordable path stacks reductions that are all now in place. First, fully implicit stepping at CFL-free time steps: backward Euler with the assembled Jacobian carries the 30 m, 2.93M-cell Harvey-region mesh at $\Delta t = 1$ s — a step the differentiable explicit path cannot take at any tested size (Section 6) — with Newton converging in about two iterations per step behind a block-Jacobi-preconditioned Krylov solve at converged inner tolerances; larger implicit steps are under investigation. Second, GPU execution of the entire gradient pipeline: the right-hand side, the assembled Jacobian and its transpose solves, the preconditioner setup, the parameter Jacobian, the observation operator, and the revolve-checkpointed trajectory are all device-resident. On four A100 GPUs, a forward-plus-adjoint gradient of a one-simulated-hour window on that mesh (3,600 implicit steps, 400 in-memory checkpoints) takes about nine minutes, and one TAO iteration of a dense-gauge twin calibration (418,076 gauges, 39-step window) takes 1.7 s — roughly $19\times$ one 64-core CPU node of the same system. Device and host runs produce bitwise-identical trajectories under a common compiler and gradients identical to every printed digit, so the finite-difference gates of Table 1 carry over unchanged. Third, land-cover-class parameterization (tens of TAO evaluations instead of hundreds) and calibration windows placed on the storm peak rather than the full event.

Red/tiger-team review: the weakest parts of this draft

[Temporary section, removed before submission. Our own adversarial read of the draft — the places a hostile reviewer (or a careful friend) will push, ranked by how much damage they could do, each stated as of today. Cross-references Qn point into the numbered issues of the next section.]

1. **There is no converged real-data *calibration* in the paper yet.** The abstract’s differentiator — production code made differentiable, versus reduced solvers rewritten in ML frameworks — invites comparison with Inunda’s calibrated 0.67 m MAE against surveyed Harvey high-water marks. What exists today is the uncalibrated baseline and its decomposition (3.41 m over all 108 marks; 1.51 m over the 71 in reaches the model drains), the identifiability twin at the real mark geometry, a bit-exactly validated mid-event restart, and — new since the last draft — an optimizer-independent measurement of how much of the

residual roughness could ever explain (Section 7.3). The first production calibration attempt took *zero* steps; the cause was a scaling defect in the optimization problem, not in the gradient, and it is diagnosed, fixed, and gated (Section 5, “Scaling the optimization problem”). A reviewer is entitled to ask whether the negative identifiability result is a property of the problem or of an optimizer we never got working; the honest answer is that the α -scan is optimizer-independent and settles it, but the argument is materially stronger with a converged run that lands where the scan predicts, and that run is in flight rather than in the paper. Until it lands, the paper claims capability, baseline, decomposition, and an authority bound — not the closed loop. (Q1, Q2)

2. **The identifiability caveats are weaker than they were, but the twins still carry them.** The measured study (Section 7.1) — 13 gauges fail, 108 marks identify the classes covering 83% of the domain — rests on a one-hour window, a 15-iteration cap and no representation error, which is exactly where a reviewer will push. Section 7.3 now answers that push from a different direction: the α and half-domain scans are on the *production* 12-hour window against *real* survey data, carry no iteration cap because no optimizer is involved, and reach the same conclusion the twins did. The remaining exposure is narrower — the twins are still the only evidence about *which* classes are recoverable, as opposed to how many. (Q2)
3. **Long-window FD gates measure the objective’s nonsmoothness, not gradient error — and the paper must not be read as claiming more.** The gradient is verified to the inner-solve floor on short windows at the production tolerances (which are now converged tolerances, closing the old verified-versus-production gap), and the long-window failure of the naive gate is measured and explained (Section 7.2). The exposure is a reviewer reading “FD-gated in CI” as covering production window lengths. (Q3)
4. **“Implicit stepping is required” means required for the differentiable right-hand side.** The operator-split friction the forward-only code uses is stable under explicit stepping but buries Δt inside the RHS; the differentiable RHS is friction-stiff ($\Delta t \lesssim 2.4$ ms on the spun-up 30 m state) and diverged at every tested explicit step. Any wall-clock comparison against operator-split explicit stepping compares different discretizations and is labeled as such. Still open: $\Delta t \geq 5$ s is blocked by a preconditioner failure. (Q6)
5. **We changed the physics to fix the solver, and the changes await domain sign-off.** The regularized drag velocity ($h_{\text{anuga}} = 0.001$ m, chosen as “smallest that cures the stall”; 0.01 m observed unstable and the mechanism only conjectured) and the transmissive outlet (replacing critical outflow; alters outlet physics; weakly reflective for subcritical flow) are both defensible and both flagged opt-in, but a reviewer can fairly ask whether the calibrated n compensates for boundary and regularization choices rather than physics. Sensitivity of the calibrated field to h_{anuga} and to the outlet choice should be demonstrated, not asserted. (Q4, Q5)
6. **Noise response is measured but only partly built into the method.** The basin-scale noisy twin shows the failure mode (misfit falls $9.4\times$ while five classes pin to bounds, Table 3). The Tikhonov weight is no longer chosen by judgment — it is σ_n^{-2} for a stated prior standard deviation on Manning n , Eq. (5) — but that only relocates the judgment into σ_n , which the domain team has not yet fixed, and the value we are using (0.015) is close to the largest that leaves the dominant direction with an interior minimum. Discrepancy-principle stopping still does not exist, and in any case the discrepancy target $J \approx N/2$ is unreachable here: the

achievable floor is set by structural error an order of magnitude larger than the 0.15 m survey error, so the floor must be reported as measured rather than assumed. (Q7)

7. **The peak-WSE argmax remains formally open.** Its kinks are measured absent at this geometry twice over — no argmax moved across any FD probe, and the production objective is bit-identical across three domain decompositions — but a remedy is proposed (frozen-argmax outer iteration) and not implemented, and the theoretical footing is the standard one for max-type observables rather than a theorem. (Q2, Q3)
8. **Reproducibility currently rests on an unmerged PETSc fork.** The device-resident assembled-Jacobian path (blocked-COO BAIJ assembly on GPU) exists only on our PETSc branch. Until it is upstreamed or the paper’s results are reproducible with stock PETSc (AIJ path, at some cost), the artifact story is weaker than the “behind a flag, FD-gated in CI” framing suggests. (Q8)
9. **The novelty claim is narrowed twice over, and what is left is not the adjoint.** “First E3SM component with built-in discrete-adjoint parameter estimation” was wrong (MALI inverts for basal friction with an AD adjoint of its steady velocity solver [18, 17]); the claim now reads first *transient* discrete adjoint in an E3SM component. A reviewer who knows MITgcm will press further and should: an exact AD adjoint of a production ocean model has driven ECCO estimation for two decades [15, 16], so “production code made differentiable” is a route with a long pedigree, not an innovation. Nor is the observable ours: Pujol et al. assimilate high-water marks variationally for distributed friction, and use adjoint sensitivity to choose the parameterization [9]. What survives is (i) the exact assembled Jacobian serving the implicit solve and the adjoint from one artifact, with the argument that implicit stepping is required for a differentiable friction RHS, (ii) basin scale on GPU, and (iii) the authority measurement of Section 7.3, which we have not seen reported for any flood-model parameter. The draft should lead with (iii). Remaining check before submission: ocean/MPAS adjoint efforts, and whether any E3SM land-model calibration uses a true adjoint rather than ensemble or surrogate gradients.

Open issues and current positions, in order of importance

[Temporary section, removed before submission. These are the open issues, in the order they matter to the paper, tagged with the team each belongs to: science (Gautam, Donghui, and colleagues), numerics (Matt, Jed), engineering (Jeff). Each states what we found, what we did, and the position we are proceeding from — we are not blocked on any of them. Nomenclature: “WSE” is water-surface elevation (bed elevation plus water depth), which is what a stage gauge reports once put on the NAVD88 datum; “cell bed” is the single mean ground elevation our 30 m mesh stores for a cell; “HWM” is a USGS-surveyed high-water mark, the peak WSE reached at a point during the event.]

1. **[Science] The real-data calibration stands on high-water marks, not stage.** The gauges are unusable at this resolution: modelled WSE sits 1–11 m above observed stage with a time-varying offset, because a 30 m cell stores the floodplain’s mean elevation while the gauge hangs in the incised channel below it (Section 7.2 has the full account and the mark QC). *Position:* peak WSE at the 108 QC-passed marks is the primary observable. Gauge stage stays available as a secondary misfit restricted to times when model and observation are

both above the cell bed, and flood-extent agreement is a candidate third; neither is needed for the headline number.

2. **[Science + numerics] Which parameterization the data actually supports.** The evidence: 238 gauge observations fit essentially perfectly while a per-cell field is 45% wrong; a class-parameterized twin at the real 13-gauge geometry (short window) drove the objective down $128\times$ with class values still 66% wrong; the same experiment at the 108 HWM locations identifies the classes covering 83% of the domain (Section 7.1). The draft’s counting argument (“overdetermined $18\times$ ”) is indefensible and is gone. Three further measurements on the production configuration settle the question quantitatively: the uniform- α scan bounds the whole roughness knob at 0.08m of a 0.72m error; the half-domain scans show the modes are not separable, each half alone making the fit worse (Section 7.3); and one BLMVM step with all fifteen classes free drives 69.7% of the domain onto or against a bound, with developed-low rising $2.6\times$ while developed-medium falls to the floor — a compensating pair of interleaved classes that is nearly null-space for a path-integrated observable. *Position:* the identifiability study — which classes are observable from which networks and windows — is a paper contribution in its own right. The production parameterization is *not* all fifteen NLCD classes: the observable supports roughly one degree of freedom, so the production run calibrates one to three parameters with the remainder frozen at the NLCD prior, and reports the fifteen-class run as the control that shows why. Freezing weakly-constrained parameters is the structural fix; leaning on the Tikhonov weight to suppress them after the fact is not.
3. **[Numerics] Inner tolerances and gradient fidelity.** Resolved by measurement: loose Krylov tolerances were a false economy (tightening $10^{-2} \rightarrow 10^{-3}$ is *faster*; the converged setting `ksp` 10^{-4} / `snes` 10^{-5} reproduces the fully converged 72-hour answer to every printed digit at $\sim 25\%$ extra cost), so campaigns now run at converged tolerances and the verified gradients are the production gradients (Section 4). The long-window FD-gate question is the objective’s nonsmoothness, not tolerance (Section 7.2). *Open:* principled handling of the remaining inexactness — forcing terms tied to line-search decrease, and nonsmooth-aware stopping for the quasi-Newton outer loop.
4. **[Science] Outflow boundary at zero velocity.** The critical-outflow condition switches abruptly at zero normal velocity: below it the boundary acts as a wall, above it it imposes critical outflow, so the flux jumps by roughly $gh^2/2$ as a boundary cell crosses stagnation, which pins the implicit solver (isolated by controls: a reflecting outlet completes the identical run; $5\times$ stronger friction regularization changes nothing; tolerance relief is a treadmill because the pin level follows the local flow state). We implemented the least clever standard treatment: a *transmissive* (zero-gradient) free outflow — ghost state equal to the interior state, as in ANUGA and Clawpack — continuous by construction, differentiated exactly through host and device Jacobians. It cures the stall: the failing control completes with zero failures and two-iteration Newton solves, and a 72-hour rain-forced forward completes 259,200 implicit solves with zero failures. It is opt-in (`type: free-outflow`); the original remains the default. Caveats: exactly non-reflecting only for supercritical outflow; does not prevent inflow if the interior velocity points inward. *Position:* transmissive outflow for the production runs, with a Froude-limited or blended form as the fallback if those caveats bite on Buffalo Bayou.
5. **[Science] Bed-friction regularization.** The friction term $\tau_b = gn^2h^{-7/3}\mathbf{q}\|\mathbf{q}\|$ switched off below the depth cutoff makes the equations jump as a cell crosses it; the jump grows like n^2 , invisible in the twins but disabling at land-cover roughness ($\bar{n} = 0.10$). We now evaluate the

friction with the ANUGA-regularized velocity, so drag fades smoothly to zero with depth. Off by default; runs use $h_{\text{anuga}} = 0.001$ m, the smallest value that cures the stall; 0.003 m is indistinguishable; 0.01 m is unstable (mechanism conjectured: weaker damping lets an iterate reach $h < 0$). *Position:* keep the ANUGA form at $h_{\text{anuga}} = 0.001$ m. The value is an empirical choice, not a physical one, and the sensitivity above is the whole of our evidence for it.

6. **[Numerics] Implicit solver headroom: bigger steps and better preconditioning.** Backward Euler with the assembled Jacobian carries the 2.93M-cell mesh at $\Delta t = 1$ s at ~ 4 –5 Newton iterations per step with `pbjacobi`-preconditioned GMRES; at $\Delta t = 5$ s the very first Newton *linear* solve diverges (300 GMRES iterations, `pbjacobi`) — a preconditioner failure before nonlinearity even enters. A wall-clock ratio against *explicit stepping of the same right-hand side* is not the modest number one might expect: with the Δt -free drag the adjoint requires, forward Euler is limited by the friction rate, $\Delta t < 1/t_b$ with $t_b = gn^2h^{-4/3}|v|$, and not by the CFL. On the spun-up 30 m state (99.9% wet, median depth 6 mm, t_b up to 420 s^{-1}) that is $\Delta t \lesssim 2.4$ ms against a CFL-limited 0.25 s — a factor ~ 400 in step count, in the implicit path’s favour. Confirmed end to end on the rain-forced window: forward Euler returns a non-finite state at $\Delta t = 0.25, 0.125$ and 0.05 s, on configurations differing from the backward-Euler run in the integrator and step size alone, while backward Euler at $\Delta t = 1$ s completes cleanly. The comparison that *would* be modest — explicit stepping at 0.25 s using the operator-split friction — is a comparison against a different, non-differentiable discretization, and we note it as such rather than quoting it as this method’s cost. *Open:* what unlocks $\Delta t \geq 5$ s. The candidates are a stronger preconditioner (AMG on a reduced system, block ILU, field-split), pseudo-transient continuation or another globalization once the linear solve holds, or simply accepting more Krylov iterations; whether to keep reassembling the Jacobian every step is open alongside it.
7. **[Science] Production-run design: window, observation error, gauge trust.** Four choices remain, two of them now settled by measurement. (a) *Settled.* Which phase of Harvey to calibrate on: the per-mark crest times decide it, and they are bimodal (hours 29–42 and 60–72) with an empty gap, so no 12-hour window covers both. The window is the upstream cluster, event hours 29–41, whose 46 marks are the only population in a reach the model drains correctly (Table 6); the middle band carries the same defect at intermediate strength and so cannot serve as held-out data either, which forces cross-validation *within* the upstream band. (b) The observation error σ per observable, which sets the data-versus-prior weight directly — USGS quality codes give 0.1–0.3 m for HWMs, and we are using 0.15 m. (c) The exclusion of eight reservoir-influenced sites (Addicks/Barker and Buffalo Bayou downstream), on the grounds that gate releases are unmodelled — now backed by a measurement on the mark side: the 37 downstream marks that never crest sit in cells the model demonstrably cannot drain (monotone filling through all 72 hours, zero recession, +7.1 m above every surveyed value; Section 7.2), so scoring or calibrating against them would fold a conveyance defect into the roughness field. (d) *Superseded.* The bounds are no longer stated in physical n : the production problem optimizes $\alpha = n/n_{\text{prior}}$ with $\alpha \in [0.3, 3]$ (Section 5), which is both a physical statement about how wrong the NLCD lookup can be and a bound that excludes the region where the implicit solve fails — a box in n did neither. The NLCD field remains the prior throughout, replacing the two-zone twins’ constant prior. *Newly open, and the most consequential of the four:* the prior width σ_n , which now sets the regularization directly through $\beta = \sigma_n^{-2}$. It is not a free knob: at $\sigma_n = 0.015$ the misfit still outbids the prior on the fifteen-class problem (the prior charged 118.5 units of objective and the misfit paid 146.5),

and the value that would make that step net-unfavourable is $\sigma_n < 0.0135$, while $\sigma_n = 0.010$ leaves the dominant direction with an optimum at $\alpha \approx 0.88$ — a well-posed problem worth almost nothing. That tension is the result, not a tuning difficulty: within a defensible prior the observable cannot support a defensible roughness field. How wrong the NLCD lookup can plausibly be is a question for the domain team, and it now has a number attached to it.

8. **[Numerics + engineering] Upstreaming the PETSc-side pieces.** The device-resident assembled-Jacobian path relies on our PETSc branch: blocked-COO assembly for BAIJ matrices on GPU (`MatCOOUseBlockIndices`; `baijkokkos/baijcuda` classes) plus small TSAdjoint fixes. We think it is generally useful — one index pair per block is the natural COO for FV systems — and intend to upstream it as is. *Open:* whether the API needs changes before it hardens, and whether the paper’s artifact claim should be scoped to the stock-PETSc AIJ path until it lands.
9. **[Engineering] Public API and CI.** The adjoint driver reaches through `private/rdycoreimpl.h` (the LETKF-driver pattern). *Open:* which calls deserve promotion to `rdycore.h` (`RDySetObservations`, `RDyAdjointSolve`, `RDyGetSensitivities` are the candidates), and whether Fortran bindings are needed at that point. The new tests need a TSAdjoint-capable PETSc and run in seconds; the CEED-only lanes correctly skip `numerics.jacobian` configs.
10. **[All] Venue and authorship.** Target journal (WRR / JAMES / GMD) and the co-author list remain to be settled.

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